INCH-POUND
MIL-DTL-24643/57F
1 October 2009
SUPERSEDING
MIL-DTL-24643/57E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LS2UW (INCLUDING VARIATION LS2UWS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS2UW and LS2UWS (double overall shield).

Construction, Watertight

First	-	Copper conductor, tin-coated, 26 AWG, 7 strands, ASTM B286.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation nominal wall thickness 0.010 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification code shall be applied by Method 3.
Fourth	-	Two conductors cabled together with a nominal lay of 1½ inches to form a pair.
Fifth	-	Forty-two pairs cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with no. 1 from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Braided shield of tin-coated copper.

Construction, Variant for Type LS2UW

Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket. Minimum average wall thickness of 0.065 inch. Overall cable diameter shall be 0.750 inch minimum, 0.790 inch maximum.

Construction, Variant for type LS2UWS

Eighth	-	Spirally-wrapped polyethylene terephthalate (polyester film) separator tape.
Ninth	-	Braided shield of tin-coated copper.
Tenth	-	An optional binder.

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Eleventh - Cross-linked polyolefin jacket. Minimum average wall thickness of 0.065 inch. Overall cable diameter shall be 0.785 inch minimum, 0.825 inch maximum.

NOTES:

1. Variant LS2UW-42 military part number M24643/57-01UO.
2. Variant LS2UWS-42 military part number M24643/57-03UD.

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	45.19
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to overall shield	1000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Overall shield to overall shield	100
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance, at 1 megahertz (MHz) (ohms)	115±15
Watertightness, leakage (in^3, max.)	1.0
Group B:	
Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

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Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Bending endurance	
At 25 °C (cycles, min.)	25
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Cable fill (sealant) removability	No failure
Shield conformance to material, construction and coverage	No failure
Surface transfer impedance (Type LS only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2

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Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2UW-42 and LS2UWS-42	1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-061)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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